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3.3 Waves

1. Progressive Waves

Progressive wave: A wave that transfers energy from one place to another without transferring matter. The oscillations of particles (or fields) propagate outward from the source. Each particle oscillates about its equilibrium position but does not travel with the wave.

Transverse vs Longitudinal

Property	Transverse Wave	Longitudinal Wave
Oscillation direction	Perpendicular to the direction of wave travel	Parallel to the direction of wave travel
Examples	EM waves, water surface waves, waves on a string, S-waves (seismic)	Sound waves, P-waves (seismic), ultrasound, compression springs
Can be polarised?	Yes	No
Features	Crests and troughs	Compressions and rarefactions
Requires a medium?	EM waves: no; mechanical: yes	Yes (always mechanical)

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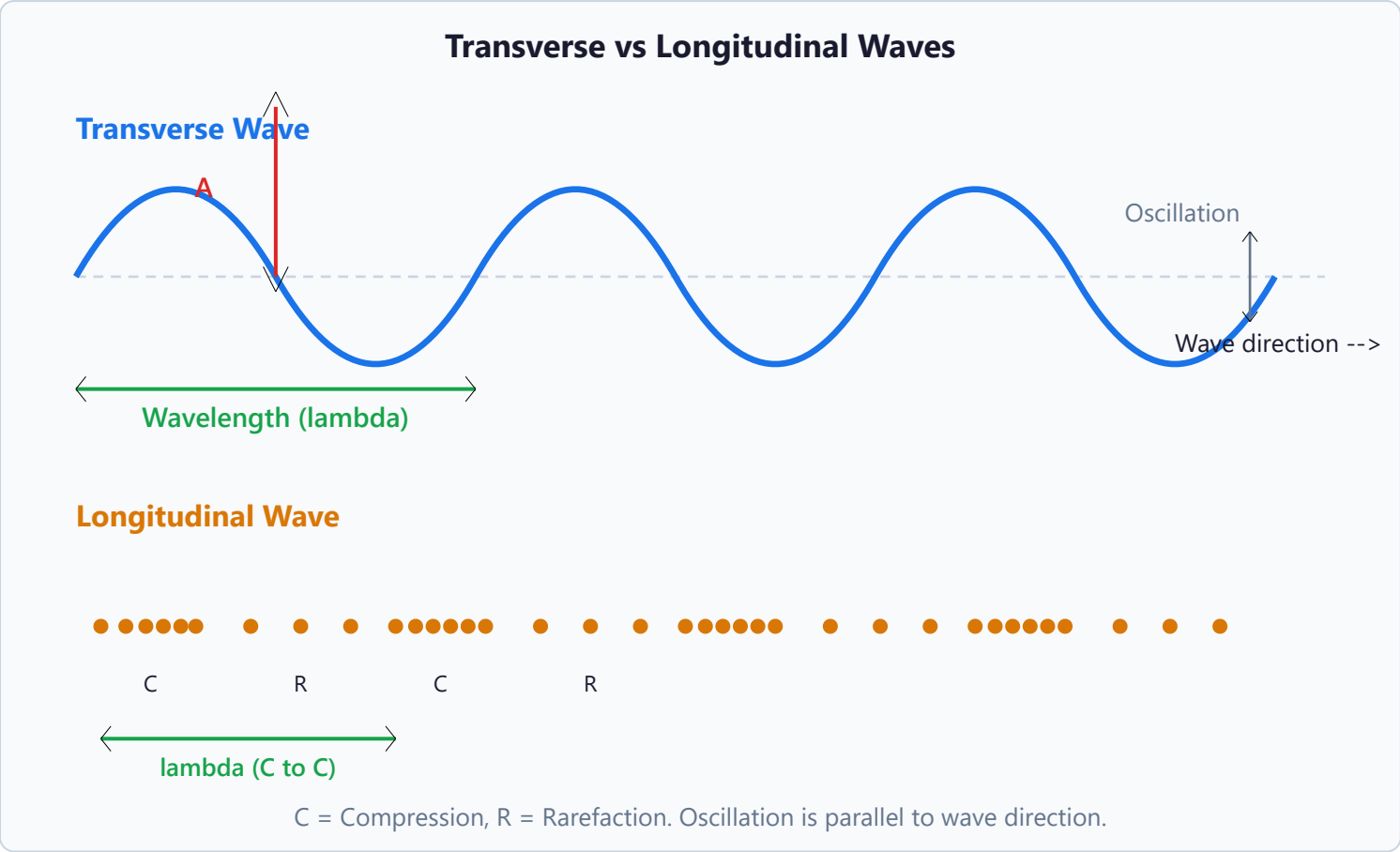


Wave Properties

- **Amplitude (A):** Maximum displacement of a particle from its equilibrium position. Related to intensity: $I \propto A^2$. Larger amplitude means more energy.
- **Wavelength (λ):** The minimum distance between two points in phase (e.g. crest to crest, compression to compression). Units: m.
- **Frequency (f):** Number of complete oscillations per second. Units: Hz (s^{-1}). Determined by the source and does not change when a wave enters a different medium.
- **Period (T):** Time for one complete oscillation. T = 1/f. Units: s.
- **Wave speed (v):** The speed at which the wave front moves through the medium. Different from the speed of individual particles.

The Wave Equation

$$v = f\lambda$$

where v = wave speed (m s^{-1}), f = frequency (Hz), λ = wavelength (m). This equation applies to all waves.

Derivation: In one period T, a wave travels exactly one wavelength λ. Speed = distance/time = λ/T = λ × (1/T) = fλ.

Worked Example: A radio station broadcasts at 98.5 MHz. Calculate the wavelength of the signal.

$v = c = 3.00 \times 10^8$ m s^{-1} (radio waves are EM waves)
$f = 98.5 \times 10^6$ Hz
$\lambda = v/f = 3.00 \times 10^8 / 98.5 \times 10^6$ = **3.05 m**

Worked Example: Sound waves of frequency 440 Hz travel at 340 m s^{-1} in air. Find the wavelength.

$\lambda = v/f = 340 / 440$ = **0.773 m** (about 77 cm)

Intensity

Intensity: The power transmitted per unit area perpendicular to the wave direction. I = P/A. Units: W m^{-2}. For a point source radiating uniformly: $I = P/(4\pi r^2)$, so intensity follows an inverse square law.

$$I \propto A^2$$

Doubling the amplitude quadruples the intensity. This relationship holds for all waves.

Phase Difference

Phase difference: The fraction of a cycle by which one wave leads or lags another. Measured in degrees (0°-360°) or radians (0-2π).

- **In phase:** Phase difference = 0 (or 2π, 4π, ..., i.e. any integer multiple of 2π). Points at the same displacement, moving in the same direction.
- **Antiphase (completely out of phase):** Phase difference = π (or 180°, or any odd multiple of π). Opposite displacements.
- **Quadrature:** Phase difference = π/2 (90°). One wave is at maximum when the other is at zero.

$$\text{Phase difference} = (2\pi / \lambda) \times \text{path difference}$$

$$\text{Phase difference (in radians)} = (2\pi / T) \times \text{time difference}$$

Worked Example: Two speakers emit sound of wavelength 0.50 m. Point P is 3.25 m from speaker 1 and 4.00 m from speaker 2. What is the phase difference and do the waves interfere constructively or destructively?

Path difference = 4.00 - 3.25 = 0.75 m
Phase difference = (2π / 0.50) × 0.75 = 3π radians = 540°
3π = π + 2π, which is equivalent to π (antiphase).
Path difference / λ = 0.75 / 0.50 = 1.5 = (1 + ½)λ
The waves arrive in **antiphase: destructive interference**.

Exam Tip: Coherence does NOT mean the waves are in phase — it means the phase difference is constant over time. Two coherent sources could have a fixed phase difference of, say, π/3. A laser is coherent; a filament lamp is not.

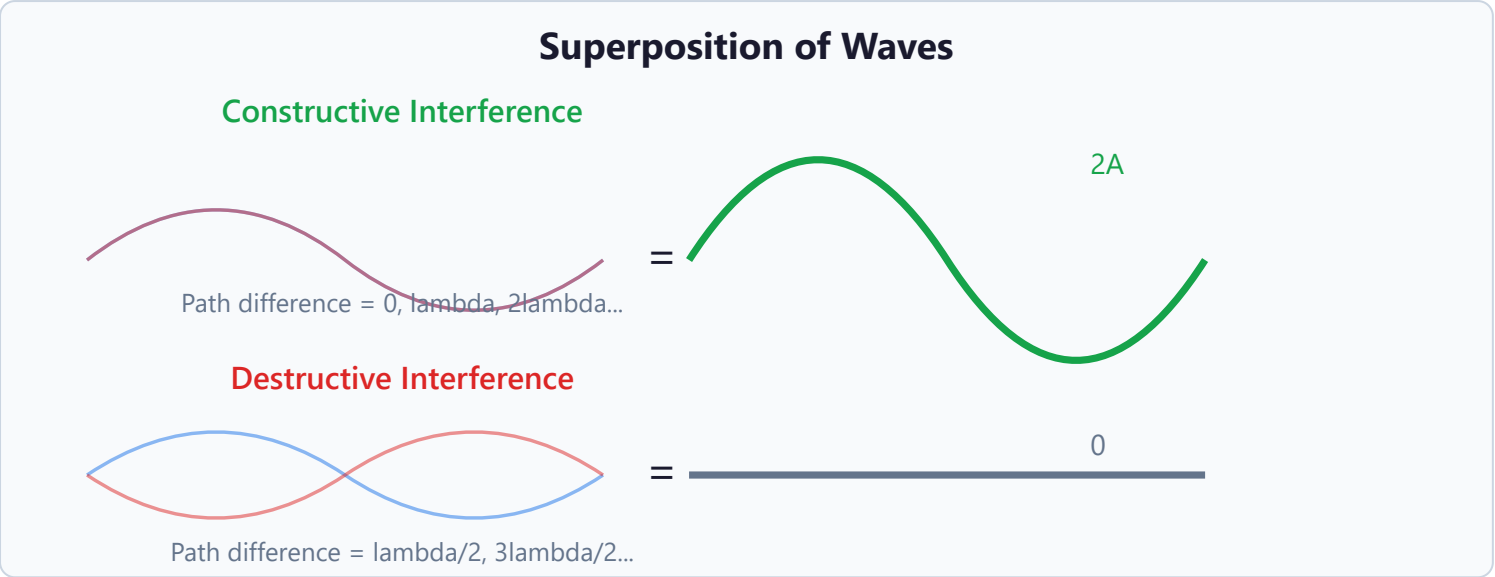
2. Superposition and Interference

Principle of superposition: When two or more waves meet at a point, the resultant displacement is the vector sum of the individual displacements at that point and at that instant.

Interference

- **Constructive interference:** Waves arrive in phase. Resultant amplitude = sum of amplitudes. Path difference = nλ (n = 0, 1, 2, ...)
- **Destructive interference:** Waves arrive in antiphase. Resultant amplitude = difference of amplitudes. Path difference = (n + ½)λ

Coherent sources: Two sources that emit waves with a constant phase difference and the same frequency. Coherence is required to observe a stable, observable interference pattern. Without coherence, the interference pattern shifts randomly and averages out.



Real-world application: Noise-cancelling headphones work by detecting ambient sound and producing a wave of equal amplitude but opposite phase (antiphase). Superposition of the two waves results in destructive interference, greatly reducing the perceived noise. Active noise cancellation is particularly effective for low-frequency, steady sounds like aircraft engine hum.

3. Young's Double Slit Experiment

Monochromatic, coherent light passes through two narrow slits separated by distance *d*. An interference pattern of bright and dark fringes is observed on a screen at distance *D*. This experiment (1801) provided the first strong evidence for the wave nature of light.

$$w = \lambda D / d$$

- w = fringe spacing (distance between adjacent bright fringes) (m)
- λ = wavelength (m)
- D = distance from slits to screen (m)
- d = slit separation (m)

Worked Example: A laser of wavelength 635 nm illuminates a double slit with separation 0.25 mm. The screen is 2.50 m away. Calculate the fringe spacing.

$w = \lambda D/d = (635 \times 10^{-9} \times 2.50) / (0.25 \times 10^{-3})$

$w = 1.5875 \times 10^{-6} / 2.5 \times 10^{-4} =$ **6.35×10^{-3} m = 6.35 mm**

White light produces a central white fringe (all wavelengths have zero path difference) with coloured fringes either side. Red fringes are wider than blue because $\lambda_{red} > \lambda_{blue}$, so $w_{red} > w_{blue}$. Away from the centre, the colours overlap and the pattern becomes less distinct.

Exam Tip: A single slit is sometimes placed before the double slit to ensure coherence when using a white light source (not needed with a laser). The single slit acts as a point source so both slits are illuminated by the same wavefront. With a laser, the light is already coherent.

4. Diffraction Gratings

A diffraction grating has many equally-spaced slits (typically hundreds per mm). It produces sharp, bright maxima at specific angles, much narrower than the double slit fringes.

$$d \sin\theta = n\lambda$$

- d = slit spacing (distance between adjacent slits) = 1 / (number of slits per metre)
- θ = angle of diffraction from the central maximum (n=0)
- n = order of the maximum (0, 1, 2, ...)
- λ = wavelength

The maximum number of orders visible is when $\sin\theta \leq 1$, so $n_{max} = d / \lambda$ (rounded down to the nearest integer).

Worked Example: A diffraction grating has 600 lines per mm. Monochromatic light of wavelength 550 nm is incident normally. Calculate (a) the slit spacing, (b) the angle of the 2nd order maximum, (c) the maximum number of orders visible.

(a) $d = 1 / (600 \times 10^3) = 1.667 \times 10^{-6}$ m

(b) $d \sin\theta = n\lambda$
$\sin\theta = n\lambda/d = (2 \times 550 \times 10^{-9}) / (1.667 \times 10^{-6}) = 0.660$
$\theta = \sin^{-1}(0.660) =$ **41.3°**

(c) $n_{max} = d/\lambda = 1.667 \times 10^{-6} / 550 \times 10^{-9} = 3.03$
$n_{max} =$ **3** (round down). So orders n = 0, 1, 2, 3 are visible on each side.

Central maximum is widest and brightest. Subsidiary maxima decrease in intensity.

Exam Tip: If given "600 lines per mm", then d = 1 / (600 × 10^3) = 1.67 × 10^{-6} m. Do NOT forget to convert mm to m. Also, gratings produce sharper, brighter maxima than the double slit because many slits contribute — this makes gratings better for precise wavelength measurement.

Gratings vs Double Slit

Double Slit	Diffraction Grating
Broad, fuzzy fringes	Sharp, narrow, bright maxima
Fringes are evenly spaced and similar intensity	Maxima are widely spaced with dark regions between
Limited precision for wavelength measurement	High precision for wavelength measurement

Real-world applications of diffraction gratings: Spectroscopy (identifying elements in stars, chemical analysis), measuring wavelengths precisely, separating white light into a spectrum (used in spectrometers), holography, and the iridescent colours on CDs/DVDs (which act as reflection gratings).

5. Diffraction

Diffraction: The spreading of waves when they pass through a gap or around an obstacle. It is most significant when the gap width is comparable to the wavelength. Diffraction is evidence of wave behaviour.

- Gap much larger than λ: almost no diffraction, wave passes through with minimal spreading.
- Gap approximately equal to λ: maximum diffraction, wave spreads out in a semicircle.
- Gap much smaller than λ: very little wave energy passes through.

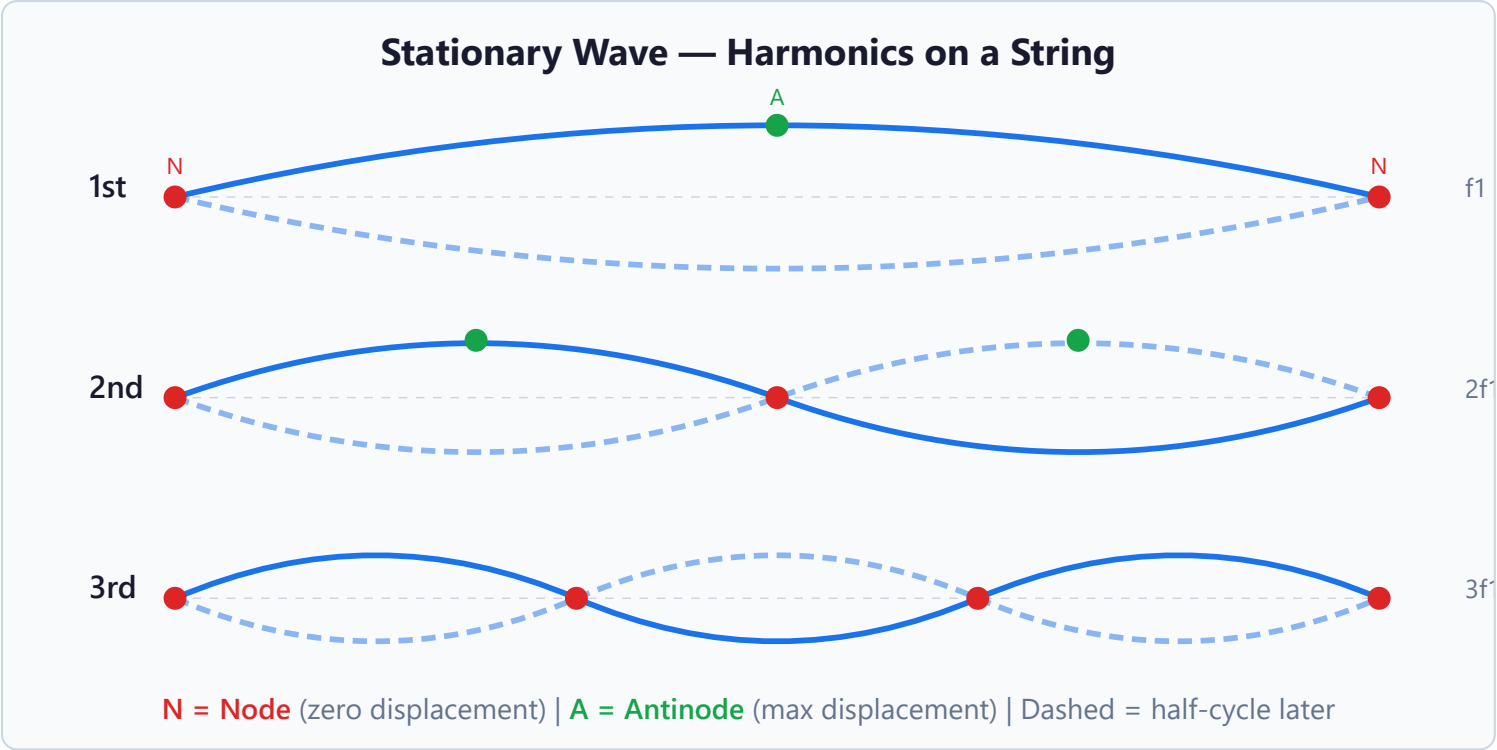
Diffraction explains why you can hear sound around corners (sound wavelengths ~1 m, similar to doorway widths) but cannot see around corners (light wavelengths ~500 nm, much smaller than everyday objects).

6. Stationary (Standing) Waves

Stationary wave: Formed by the superposition of two progressive waves of the same frequency, same amplitude, and same wavelength, travelling in opposite directions. Energy is stored between the nodes, not transferred along the wave. Stationary waves do not propagate.

Nodes and Antinodes

- **Node:** A point of zero displacement (always stationary). Formed by permanent destructive interference.
- **Antinode:** A point of maximum displacement. Formed by constructive interference.
- The distance between adjacent nodes (or adjacent antinodes) = λ/2.
- All points between two adjacent nodes oscillate in phase with each other.
- Points on opposite sides of a node are in antiphase (π phase difference).



Harmonics on a String

Harmonic	Name	Nodes	Antinodes	Wavelength	Frequency
1st	Fundamental	2	1	$\lambda_1 = 2L$	f_1
2nd	1st overtone	3	2	$\lambda_2 = L$	$2f_1$
3rd	2nd overtone	4	3	$\lambda_3 = 2L/3$	$3f_1$
nth	—	n+1	n	$\lambda_n = 2L/n$	nf_1

The fundamental frequency of a vibrating string depends on its length, tension, and mass per unit length:

$$f_1 = (1 / 2L) \sqrt{(T / \mu)}$$

where L = length, T = tension (N), μ = mass per unit length (kg m^{-1}).

Worked Example: A guitar string of length 0.65 m and mass per unit length 3.5×10^{-3} kg m^{-1} is under a tension of 100 N. Find the fundamental frequency and the frequency of the 3rd harmonic.

$f_1 = (1 / 2L)\sqrt{(T/\mu)} = (1 / (2 \times 0.65))\sqrt{(100 / 3.5 \times 10^{-3})}$

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= (1/1.30) × √(28571) = 0.769 × 169.0 = **130 Hz**

3rd harmonic: $f_3 = 3f_1 = 3 \times 130$ = **390 Hz**

Stationary Waves in Pipes

Open pipe (both ends open): Antinodes at both ends. All harmonics present. Fundamental: $\lambda = 2L$, so $f_1 = v/(2L)$.

Closed pipe (one end closed): Node at the closed end, antinode at the open end. Only **odd** harmonics are present (1st, 3rd, 5th...). Fundamental: $\lambda = 4L$, so $f_1 = v/(4L)$.

Real-world application: Musical instruments exploit stationary waves. A flute is an open pipe (all harmonics). A clarinet behaves as a closed pipe (odd harmonics only), giving it a distinctively different timbre. Organ pipes can be either open or closed.

Differences: Stationary vs Progressive

Stationary Wave	Progressive Wave
No net energy transfer	Energy is transferred in the direction of propagation
Amplitude varies with position (zero at nodes, max at antinodes)	All points have the same amplitude
Points between adjacent nodes oscillate in phase	Phase changes continuously along the wave
Frequency is fixed (only specific harmonics)	Any frequency is possible
Wavelength is determined by the boundary conditions	Wavelength depends on speed and frequency

7. Refraction

Refraction: The change in direction of a wave as it passes from one medium to another, caused by a change in wave speed. The frequency remains constant while the wavelength and speed change.

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Refractive Index

$$n = c / v$$

where c = speed of light in a vacuum (3.00×10^8 m s^{-1}), v = speed of light in the medium. The refractive index is always ≥ 1 (it equals 1 for a vacuum, ~1.0003 for air, ~1.5 for glass, ~1.33 for water, ~2.42 for diamond).

Snell's Law

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

where θ is the angle measured from the normal. When light enters a denser medium (higher n), it slows down and bends towards the normal.

Worked Example: Light passes from air (n = 1.00) into glass (n = 1.52) at an angle of incidence of 40°. Find the angle of refraction.

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$1.00 \times \sin 40° = 1.52 \times \sin\theta_2$

$\sin\theta_2 = \sin 40° / 1.52 = 0.6428 / 1.52 = 0.4229$

$\theta_2 = \sin^{-1}(0.4229) =$ **25.0°**

The light bends towards the normal (as expected, entering a denser medium).

Total Internal Reflection (TIR)

Total internal reflection: Occurs when light travelling in a denser medium hits the boundary with a less dense medium at an angle greater than the critical angle. All light is reflected back into the denser medium — no refracted ray exists.

Conditions for TIR:

1. Light must travel from a more optically dense medium to a less optically dense medium ($n_1 > n_2$).
2. The angle of incidence must be greater than the critical angle ($\theta > \theta_c$).

$$\sin\theta_c = n_2 / n_1$$

For a medium surrounded by air ($n_2 \approx 1$): $\sin\theta_c = 1/n$

Worked Example: Calculate the critical angle for light in glass (n = 1.52) meeting an air boundary.

$\sin\theta_c = n_{air}/n_{glass} = 1.00/1.52 = 0.658$

$\theta_c = \sin^{-1}(0.658) =$ **41.1°**

Any ray hitting the glass-air boundary at more than 41.1° from the normal will be totally internally reflected.

Real-world applications of TIR:

- **Optical fibres:** Used in telecommunications (internet data), medical endoscopes, and decorative lighting. The core has a higher n than the cladding, ensuring TIR keeps light trapped in the core.
- **Cladding functions:** Protects the core from damage, prevents signal leakage (crosstalk) between adjacent fibres, and maintains the conditions for TIR.
- **Signal degradation:** Absorption (energy loss to fibre material), dispersion (pulse broadening due to multipath or material dispersion), and scattering at impurities.
- **Prisms in binoculars:** Use TIR instead of mirrors (more efficient, no silvered coating needed).

Exam Tip: In optical fibres, the core has a higher refractive index than the cladding. Without cladding, any contact with another surface could allow light to escape. Signal degradation in fibres is caused by absorption (solved by using very pure glass), modal dispersion (solved by using a narrow core), and material dispersion (solved by using monochromatic light).

8. Polarisation

Polarisation: The restriction of the oscillations of a transverse wave to a single plane. Only transverse waves can be polarised — this is definitive evidence that light is a transverse wave. Longitudinal waves cannot be polarised.

- **Unpolarised light:** Oscillations occur in all planes perpendicular to the direction of travel.
- **Plane-polarised light:** Oscillations occur in only one plane.
- A **polarising filter (Polaroid)** transmits only the component of oscillation parallel to its transmission axis.

Two Polarising Filters

If two polarising filters are aligned (transmission axes parallel), maximum light passes through. If they are crossed (at 90°), no light passes through. At an angle θ between the transmission axes:

$$I = I_0 \cos^2\theta$$

This is Malus's Law, where I_0 is the intensity after the first polariser and θ is the angle between the polariser and analyser.

Worked Example: Unpolarised light of intensity I_0 passes through two polarising filters with their axes at 30° to each other. What fraction of the original intensity is transmitted?

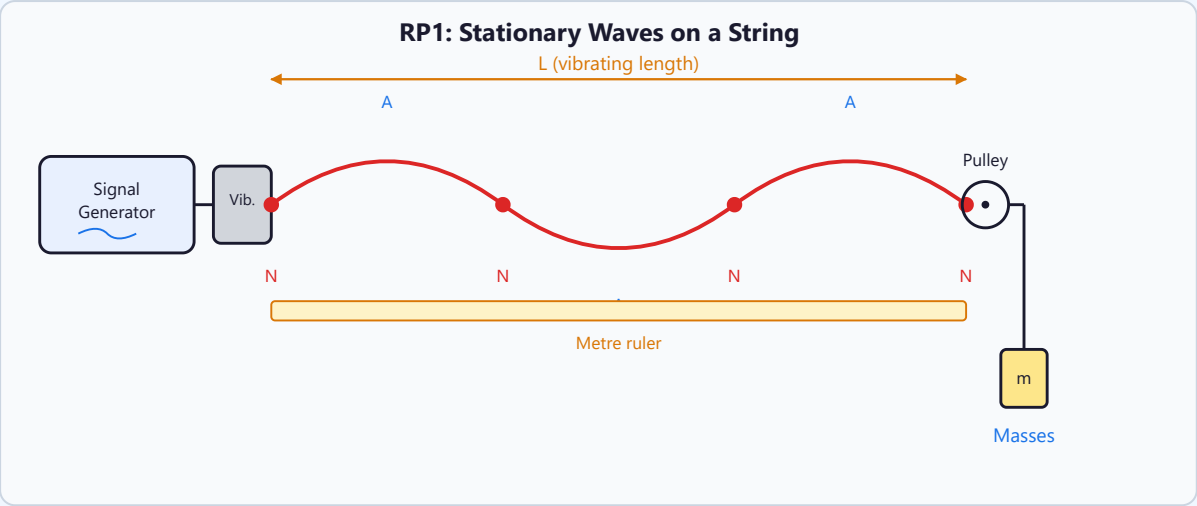
After the first filter: $I_1 = I_0/2$ (unpolarised light loses half its intensity)
After the second filter: $I_2 = I_1 \cos^2 30° = (I_0/2) \times (\sqrt{3}/2)^2 = (I_0/2) \times (3/4)$
$I_2 = \mathbf{3I_0/8 = 0.375\ I_0}$ (37.5% of the original intensity)

Applications of polarisation:

- **Polaroid sunglasses:** Reduce glare from reflected light (light reflected off water/roads is partially polarised horizontally; vertical polarising filters block this).
- **LCD screens:** Use polarising filters and liquid crystals to control light transmission pixel by pixel.
- **Stress analysis:** Transparent materials under stress rotate the plane of polarisation. Viewed between crossed polarisers, stress concentrations show as coloured fringes.
- **3D cinema:** Two images projected with different polarisations; glasses with different polarising filters for each eye.

Exam Tip: The fact that light can be polarised proves it is a transverse wave. Sound cannot be polarised because it is longitudinal. If asked for evidence that electromagnetic waves are transverse, cite polarisation experiments with Polaroid filters or microwave polarisation experiments with a metal grid.

⚗ Required Practical 1: Investigation into Stationary Waves on a String



Apparatus

- Signal generator and vibration generator (mechanical oscillator)
- Thin string or elastic cord
- Pulley and clamp, retort stand
- Slotted masses and hanger (to provide tension T = mg)
- Metre ruler

Method

1. Clamp the vibration generator to the bench. Attach the string to the vibrator, pass it over the pulley, and hang a known mass to provide tension T = mg.
2. Set a low frequency and slowly increase until the fundamental mode (one antinode/loop) is clearly visible. Record this resonant frequency f_1.
3. Measure the vibrating length L (vibrator blade to pulley contact) with the metre ruler.
4. Continue increasing frequency to find the 2nd, 3rd, 4th harmonics. Record each resonant frequency and count the number of loops n.
5. **To investigate f vs 1/L:** Keep mass (tension) constant. Change L by moving the pulley. Find the fundamental frequency for each L. Plot f against 1/L — expect a straight line through the origin.
6. **To investigate f vs tension:** Keep L constant. Change the hanging mass. Find the fundamental frequency for each mass. Plot f against √T — expect a straight line through the origin.
7. Repeat each measurement at least 3 times and calculate a mean.

Variables

Variable	f vs 1/L investigation	f vs T investigation
Independent	Length L	Tension T (hanging mass)
Dependent	Resonant frequency f	Resonant frequency f
Control	Tension, string type (μ)	Length, string type (μ)

Expected Results

$f = (1/2L)\sqrt{(T/\mu)}$, so:

- f vs 1/L: straight line through origin with gradient = $(1/2)\sqrt{(T/\mu)}$
- f vs √T: straight line through origin with gradient = $1/(2L\sqrt{\mu})$
- From the gradient, μ can be calculated and compared with a direct measurement (weigh a known length of string).

Safety

- Wear safety goggles — the string may snap under tension.
- Place a cushion beneath the hanging masses to catch them if the string breaks.
- Keep masses low to avoid string breakage.

Sources of Error and Improvements

Source of Error	Type	Improvement
Difficult to judge exact node positions by eye	Systematic	Use a strobe light; use paper riders that are thrown off at antinodes
String may not vibrate in a perfect plane	Random	Ensure string and vibrator are properly aligned
Measuring L is imprecise at the vibrator end	Systematic	Measure from node to node and count half-wavelengths
Frequency reading on signal generator may be inaccurate	Systematic	Calibrate with a digital frequency meter

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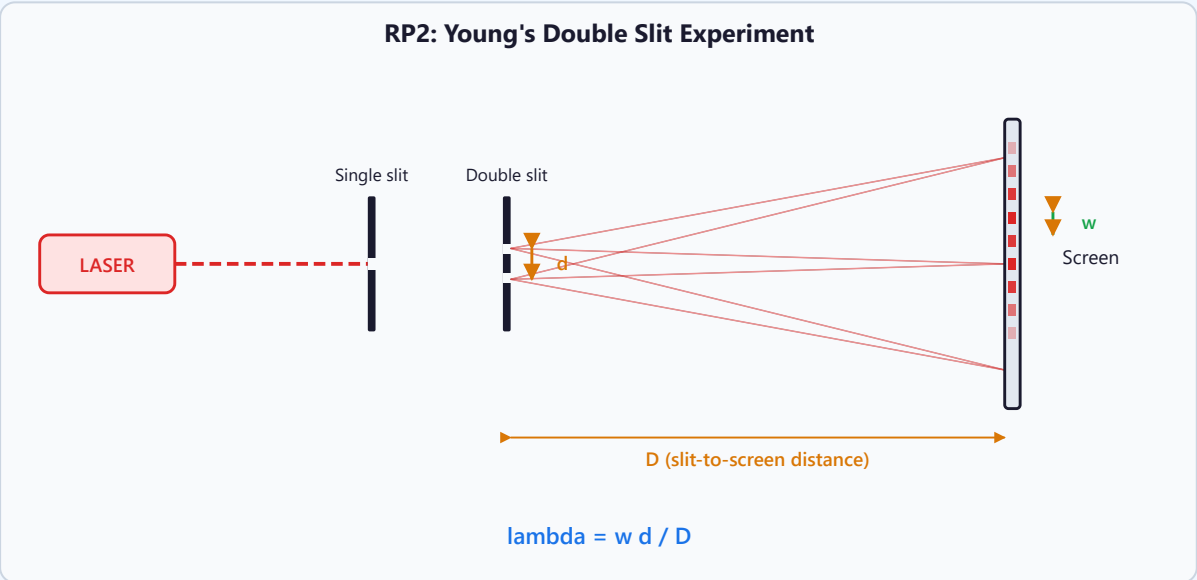
Tension changes as string forms loops (stretching)	Systematic	Use a stiff, inextensible string

Exam Tip: At resonance, the amplitude is maximum and the standing wave pattern is clearly visible. A paper rider at an antinode will be thrown off, confirming its position. A paper rider at a node remains stationary.

⚗ Required Practical 2: Investigation of Interference Effects (Young's Double Slit and Diffraction Grating)



Apparatus

- Laser (e.g. He-Ne or diode laser, known wavelength for verification)
- Double slit (known separation d, typically 0.1-0.5 mm)
- Diffraction grating (known number of lines per mm, e.g. 300 lines/mm)
- White screen, metre ruler and/or tape measure, darkened room
- Vernier callipers or travelling microscope (for measuring fringe spacing)

Method — Part A: Young's Double Slit

1. Set up the laser so the beam passes through the double slit. An interference pattern of bright and dark fringes appears on the distant screen.
2. Measure distance D from slits to screen using a tape measure (D should be large, e.g. 1-2 m).
3. Measure the fringe spacing w: measure across several fringes (e.g. 10 bright fringes) and divide by the number of fringe spacings to get an accurate mean.
4. Record the slit separation d (from the slide labelling or measured with a travelling microscope).
5. Calculate wavelength: $\lambda = wd/D$. Repeat for different D values to check consistency.

Method — Part B: Diffraction Grating

1. Replace the double slit with a diffraction grating. Observe sharp, bright maxima.
2. Measure the angle θ to each order maximum (n = 1, 2, 3...) using $\tan\theta = x/D$ where x is the distance from central maximum.
3. Apply $d \sin\theta = n\lambda$ to calculate λ for each order. Take a mean.
4. Compare with the known laser wavelength to assess accuracy.

Safety

- **Laser safety is critical:** Never look directly into the laser beam or its reflections.
- Display a laser safety warning sign on the door.
- Turn off the laser when not taking measurements.
- Remove reflective objects (watches, jewellery) from the vicinity.
- Ensure the beam path is below or above eye level.

Sources of Error and Improvements

Source of Error	Type	Improvement
Fringe spacing is small and hard to measure	Random	Measure across many fringes (e.g. 10) and divide; use a travelling microscope
D measurement inaccurate if screen not perpendicular	Systematic	Use a set square; measure D carefully to the slit plane
Central maximum incorrectly identified	Systematic	The central fringe is the brightest; ensure laser hits slit centre

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Slit separation d may be inaccurate	Systematic	Use manufacturer's value or measure with a travelling microscope

Exam Tip: The diffraction grating produces sharper, brighter maxima than the double slit because many slits contribute to constructive interference. This gives a more precise wavelength measurement. Always measure across multiple fringes to reduce percentage uncertainty in w. The grating method is better when high accuracy is needed.

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